Week 19 — Extended Reading: Preference objectives and their assumptions

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Rafailov et al. (2023): DPO

Primary source. https://arxiv.org/abs/2305.18290

Mechanism and scope. DPO substitutes the optimal KL-regularised policy's log ratio into a pairwise preference likelihood. The derivation needs a supported reference distribution and a stated preference model. The resulting training objective does not certify deployment quality.

Read, derive, investigate. Derive the policy normaliser and show why it cancels within a same-prompt pair. At policy equals reference, verify a zero logit and loss $\log 2$. Identify what happens to this statement if a nonzero margin is added.

2 Azar et al. (2023/2024): IPO

Primary source. https://arxiv.org/abs/2310.12036

Mechanism and scope. IPO emerges from a broader preference-optimisation framework and uses a squared target on a log-ratio difference. Its finite target differs from a logistic objective that can keep rewarding a larger margin on separable data.

Read, derive, investigate. Differentiate $(h - 1/(2\beta))^2$ with respect to h. Compare its zero with the DPO logistic derivative. IPO is not obtained simply by taking DPO's temperature to infinity.

3 Ethayarajh et al. (2024): KTO

Primary source. https://arxiv.org/abs/2402.01306

Mechanism and scope. KTO uses desirable/undesirable feedback rather than requiring matched preference pairs. Its prospect-theoretic construction compares a policy/reference log ratio with a reference point, with weights for the two label types.

Read, derive, investigate. Declare whether $\ell = \log \pi - \log \pi_{\text{ref}}$ already contains temperature. Apply β once, not twice. The workbook's clipped batch mean is a small reference-point proxy; distinguish it from an exact KL expectation and from the paper's estimator construction.

4 Hong et al. (2024): ORPO

Primary source. https://arxiv.org/abs/2403.07691

Mechanism and scope. ORPO combines a chosen-response likelihood objective with an odds-ratio preference term. The paper's sequence score is based on average token log likelihood; it is not the ordinary probability of a complete sequence.

Read, derive, investigate. Set $p = \exp(\text{mean token log probability})$ and compute its log odds. Differentiate to see the $1/[p(1-p)]$ sensitivity near the boundaries. Stable log arithmetic prevents cancellation but does not eliminate large mathematical gradients.

5 Meng, Xia and Chen (2024): SimPO

Primary source. https://arxiv.org/abs/2405.14734

Mechanism and scope. SimPO uses a length-normalised log-likelihood reward and a target reward-margin offset without a separate reference model. Its loss still decreases as an already favourable margin increases; the offset does not create a finite minimiser by itself.

Read, derive, investigate. Compare sum-token and mean-token scores on responses of different lengths. Distinguish the training objective from length-controlled evaluation and from the decoding strategy. Normalising length does not guarantee that all length-related preferences disappear.

6 Liu et al. (2024): Length-regularised DPO

Primary source. https://arxiv.org/abs/2406.11817

Mechanism and scope. This work studies iterative length regularisation in preference optimisation. It motivates checking whether improvement tracks task quality or a length cue in the labels and model scores.

Read, derive, investigate. Hold semantic content fixed while changing verbosity. Does the preference margin move? Report response length alongside a quality measure, and state how the regularisation changes the objective. Do not equate lower length correlation with proof of better reasoning.

7 Xu et al. (2024): Contrastive Preference Optimisation

Primary source. https://arxiv.org/abs/2401.08417

Mechanism and scope. This CPO is a preference method studied for machine translation. It is distinct from Constrained Policy Optimization in Week 22, despite the shared acronym.

Read, derive, investigate. Create a two-column glossary for the two CPOs: data, objective and constraint meaning. Identify the role of a chosen-response likelihood term in the translation method. A shared acronym does not imply shared guarantees.

8 Yuan et al. (2024): Self-Rewarding Language Models

Primary source. https://arxiv.org/abs/2401.10020

Mechanism and scope. The paper investigates iterative generation of judgments and preference training. A model-generated judge creates a feedback loop that should be evaluated against independent criteria.

Read, derive, investigate. Draw which checkpoint generates responses, which judges them and which is updated. Explain how self-consistency can increase while an external quality measure stagnates. This week's toy pairwise classifiers do not reproduce iterative self-rewarding LLM experiments.

Connect the reading to your workbook

For each objective, write a function signature listing label type, reference scores, temperature, margin and length convention. Then compare losses on identical synthetic inputs. Equal loss values across different objectives are not comparable units of human preference quality; use held-out evaluations and inspect failure modes.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.